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Bio_303

Current Mid_Term Papers

Fall 2022_23 (Jan 2_14)

Most Repeated Questions
Full Solution

Date of Collection: 11/01/2023

ZOOLOGY
LEGENDS

"Don't Stop When you Tired, Stop When you Done"

0304-4756496

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(There is Little Possibility of Any Mistake, if you find Let us to Know)

(All Ans Are Taken From PPTs)

MCQS:

- ✓ .....are single strands globular molecules.___ **tRNA**
- ✓ Animals convert the chemical energy of fuels into **motion and heat** and In a few organisms such as fireflies chemical energy is converted into **light**
- ✓ A ligand bind at a site on the protein called the___**binding site.**
- ✓ Gluconeogenesis is a process_____**Synthesis glucose**
- ✓ Cori cycle take place___**liver**
- ✓ Conversion of glucose 6_phosphate of fructose 6_ phosphate requires enzymes_____**phosphohexose isomerase**
- ✓ Standard reduction is denoted by____**E°**
- ✓ Pyruvate is oxidized, with loss of its carboxyl group as **CO2** , to yield the acetyl group of **acetyl-coenzyme A;**

- ✓ Chemicals reaction in which the reactions and products same phase____**homologous chemical reactions**
- ✓ Covalent bond are also called____**molecular bond**
- ✓ It is in the form of a loop and carries specific sequences of **three bases** which constitute the anticodon.
- ✓ Secondary structure involves **coil formation** of the polyribonucleotide chain.
- ✓ It is defined as the number and sequence of ribonucleotides in the RNA chain._____ **Primary Structure of RNA**
- ✓ Separation of the two strands of the double helix when hydrogen bonds between the paired bases are disrupted_____ **DNA Denaturation**
- ✓ In **non-oxidative phase**, ribulose 5-phosphate is first epimerized to xylulose 5- phosphate
- ✓ In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by **PEP carboxykinase.**
- ✓ The important precursors of glucose in animals are **three-carbon compounds** such as lactate, pyruvate, and glycerol, as well as certain amino acids

- ✓ We take the example of Aspergillus niger and Lactobacillus are the microbes used to commercially produce **citric acid and lactic acid**, respectively. **(Both Fermentation)**
- ✓ Lactic acid fermentation is done by some **fungi, some bacteria** like in yogurt, and sometimes by our muscles.
- ✓ Compounds with two phosphates linked together as a pyrophosphoryl group are named **diphosphates;**
- ✓ Compounds that contain two phosphate or phosphoryl groups attached at different positions in the molecule are named **bisphosphates (or bisphospho compounds);**
- ✓ The energy gain comes in the **payoff phase of glycolysis**
- ✓ Note that during preparatory phase, **2 molecules** of ATP are invested before the cleavage of glucose into two three-carbon pieces.
- ✓ **NAD and NADP** are soluble electron carriers that act with dehydrogenase enzymes
- ✓ Waste products from other industrial processes, such as molasses, cheese whey and corn steep liquor, after modifying with the incorporation of additional nutrients, are used **as the substrate** for many industrial fermentation

- ✓ • A high level of **AMP** indicates that the energy charge is low
- ✓ High levels of **ATP and citrate** indicate that the energy charge is high
- ✓ In 1910 Chaim Weizmann (later to become the first president of Israel) discovered that the bacterium Clostridium acetobutyricum ferments starch to **butanol and acetone.**
- ✓ Biological Reactions required energy ____**endergonic.**
- ✓ Actual free-Energy changes depend on **Reactant and Product** Concentrations
- ✓ The binding of a protein and ligand is often coupled to a **conformational change** in the protein
- ✓ All tRNA molecules contain **4** main arms or loops.
- ✓ Most biochemical reactions occur near **pH 7** (buffered)
- ✓ beer and wine, alcohol are produced by fermentation
- ✓ In some plant tissues and in certain invertebrates, protists, and microorganisms such as yeast, **pyruvate** is converted under hypoxic or anaerobic conditions to

ethanol and CO2, a process called ethanol (alcohol) fermentation

- ✓ Under **Aerobic conditions,** the pyruvate formed in the final step of **glycolysis** is oxidized to acetate (acetyl- CoA), which enters the citric acid cycle and is oxidized to CO2 and H2O.
- ✓ Dehydrogenase meaning ____**Removal of Hydrogen**
- ✓ **Galactokinase** deficiency **galactosemia** high conc. are found in blood and urine.
- ✓ Messenger RNA (mRNA) Size **500-6000 nucleotides**
- ✓ The rate of **re-association** depends upon the concentration of the **complementary strands**.
- ✓ **Pyruvate** is the **end product** of the second phase of glycolysis
- ✓ Tri carbocyclic acid cycle (TCA) or **Kreb's cycle**
- ✓ Kreb's cycle same as **Acetyl Co-A**
- ✓ The enzyme **phosphohexose isomerase** (phosphoglucose isomerase) catalyzes the reversible isomerization of

glucose 6- phosphate, an aldose, to fructose 6-phosphate, a ketose.

- ✓ Unshared electrons are represented by **Dot**.
- ✓ Similar to DNA, RNA polymer also has **polarity**
- ✓ Too much alcohol drinking ◊ reduce **niacin** absorption from the **intestine**
- ✓ **Oxalo acetic acid (OAA)** produced in **mitochondria** is impermeable through mitochondrial membrane
- ✓ **Lipid-soluble quinones**: ubiquinone and plastoquinone act as **electron carriers** and proton donors in the non-aqueous environment of membranes
- ✓ **NAD** is a lipid soluble electro carriers.
- ✓ In the **surface techniques**, the microorganisms are cultivated on **the surface of a liquid** or solid substrate.
- ✓ In the **submersion processes**, the microorganisms grow **in a liquid medium** provided with the substrate
- ✓ **Lyases** are included in a separate class because they catalyze these reactions by means other than **hydrolysis or oxidation.**

- ✓ **Nucleophiles:** functional groups rich in and capable of donating electrons
- ✓ The **electron-donating** molecule in an oxidation-reduction reaction is called the **reducing agent or reductant**
- ✓ The electron-**accepting** molecule is the **oxidizing agent or oxidant**.
- ✓ **Photosynthetic organisms** make glucose by first reducing atmospheric **CO2 to trioses**, then converting **the trioses to glucose**.
- ✓ **NAD, NADP** are water soluable coenzymes.
- ✓ The payoff phase of glycolysis includes the energy-conserving **phosphorylation steps**
- ✓ RNA typically exists as a **single strand Molecules** whereas DNA exists as a **double-stranded** helical molecule.
- ✓ The process of synthesis of **glucose from non-carbohydrate** sources is known as **Gluconeogenesis** (abbreviated GNG)

✓ **Which is correct about Exponential Phase?**

1)The second phase in the fermentation process

2) The cells have adjusted to their environment and rapid growth takes place

3) Cell growth rate is highest in this phase

4) All

✓ This reannealing process is also referred to as **hybridization**.

✓ Since the RNA molecule is a **single strand complementary** to only one of the two strands of a gene,

✓ miRNA is a type of **ncRNA**

✓ When the amount of ATP is high in cell, glucose is converted into **Glycogen.**

✓ Reactions in which system gains free energy is called **Endergonic.**

✓ Phosphoenolpyruvate is an **Enol Form**.

✓ A phosphoryl group (-PO3 2-) from ATP is transferred to an alcohol (◊ forming a **phosphate ester)**

ZOOLOGY LEGENDS

Short Questions:

(These are Most Repeated Questions)

✓ Types of RNA

(i)**Messenger RNAs (mRNAs)-** transfer genetic information from DNA to the protein synthesizing machinery.

(ii) Ribosomal RNAs (rRNAs)- contribute to the formation and function of ribosomes

(iii) Transfer RNAs (tRNAs)- adapter molecules that carry specific amino acids for protein synthesis

(iv) Small nuclear RNA (snRNA)- play pivotal roles in RNA processing, particularly mRNA processing

✓ Stationary phase in fermentation

This is the third phase in the fermentation process. The cell growth rate has leveled off and become constant. The number of cells multiplying equals the number of cells dying

✓ 2 names of Glycogenic amino acids

Glycine, serine, aspartic acid, glutamic acid, glutamine, valine, methionine, histidine, and arginine are all glucogenic amino acids.

✓ What is acyl group? write its formation

• It contains a double bonded oxygen atom and an alkyl group. (R-C=O group is called Acyl Group). In organic chemistry, the acyl group (IUPAC name: alkanoyl) is usually derived from a carboxylic acid. Acyl group is electron withdrawing due to the electronegativity of the oxygen atom.

✓ Continuously fermentation process

• A Continuous fermentation is a process in which fresh medium (substrate) is continuously added in the bioreactor , and biomass or products containing left over nutrients and microorganisms are continuously removed at the same rate to keep the culture volume constant.

• Under these conditions the cells remain in the logarithmic phase of growth

✓ Submersions fermentation

• In the submersion processes, the microorganisms grow in a liquid medium provided with the substrate • Except in traditional beer and wine fermentation, the medium is held in fermenters and stirred to obtain a homogeneous distribution of cells and medium.

• Most processes are aerobic, and for these the medium must be vigorously aerated.

• Used for pickling vegetables and producing soy sauce. All important industrial processes (production of biomass and protein, antibiotics, enzymes are carried out by submersion processes

✓ Define upper steam processing

Upstream processing includes formulation of the fermentation medium, sterilization of air, fermentation medium and the fermenter, inoculum preparation and inoculation of the medium.

The fermentation medium should contain an energy source, a carbon source, a nitrogen source and micronutrients required for the growth of the microorganism along with water and oxygen, if necessary.

✓ Write down the three steps of enzyme catalyzed reaction

A typical enzymatic process includes

1- **substrate binding,**
2- **a chemical step involving covalent bond rearrangements, and**
3- **product release**.

✓ Difference between pentose sugar of RNA and DNA

The pentose sugar in DNA is called deoxyribose, and in RNA, the sugar is ribose. The difference between the sugars is **the presence of the hydroxyl group on the 2' carbon of the ribose and its absence on the 2' carbon of the deoxyribose**.

✓ Reduce equivalents and its examples?

Reducing equivalent refers to **chemical species which transfer the equivalent of one electron in redox reactions**. The term is common in biochemistry. A reducing equivalent can be an electron, a hydrogen atom, as a hydride ion.

Examples of reducing equivalents are:

A lone electron

A hydrogen atom

A hydride ion ($:H^-$)0

✓ GLUT family

• Glucose uptake from the blood is mediated by the GLUT family of glucose transporters • The transporters of hepatocytes (GLUT1, GLUT2) and of brain neurons (GLUT3) are always present in plasma membranes.

• In contrast, the main glucose transporter in the cells of skeletal muscle, cardiac muscle, and adipose tissue (GLUT4) is sequestered in small intracellular vesicles and moves into the plasma membrane only in response to an insulin signal.

✓ Difference between Lyases and lygases? (3)

Lyases	Ligases
Nonhydrolytic removal of a group or addition of a group to a substrate.	(Synthetases) Joining of two molecules by the formation of new bonds.
Example: Decarboxylases, Aldolases	**Example:** Citric acid Synthetase

✓ Components/ Factors of fermenter

Factors to consider when designing a fermenter

• Aseptic and long-term reliability

• Adequate for aeration and agitation

• Low power consumption

• Temperature and pH controls

• Sampling facilities

• drain or overflow

• control systems

• Sensors

• Cooling to achieve maximum microbial yield

✓ What is Ligands? Or Characteristic of ligands?

In biochemistry and pharmacology, a ligand (from the Latin ligandum, binding) is a substance (usually a small molecule), that forms a complex with a biomolecule to serve a biological purpose.

• In a narrower sense, it is a signal triggering molecule, binding to a site on a target protein • A molecule bound reversibly by a protein is called a ligand.

• Ligands include substrates, inhibitors, activators, and neurotransmitters

✓ Organs where Gluconeogenesis occurs

• Mainly in the liver

• To a lesser extent in renal cortex

• In the epithelial cells that line the inside of the small intestine

• In cytosol and mitochondria depending upon the substrate

✓ DNA / RNA diff

- ✓ DNA contains the sugar deoxyribose, while RNA contains the sugar ribose. ...
- ✓ DNA is a double-stranded molecule, while RNA is a single-stranded molecule.
- ✓ DNA is stable under alkaline conditions, while RNA is not stable.
- ✓ Instead of thymine in DNA, RNA contains the ribonucleotide of uracil.
- ✓ RNA is considerably smaller than DNA
- ✓ RNA has no simple, structural hierarchy that serves as a reference point, as does the double helix for DNA.

✓ Cytochrome

Which have tightly bound prosthetic groups that undergo reversible oxidation and reduction, serve as electron carriers in many oxidation-reduction reactions

✓ Describe the function of phosphofructokinase-1 (PKF-1). What they are produced?

Phosphofructokinase-1 (PFK-1) catalyzes the transfer of a phosphoryl group from ATP to fructose 6-phosphate to yield fructose 1,6-bisphosphate.

KEY CONVENTION:

• The PFK-1 reaction is essentially irreversible under cellular conditions, and it is the first "committed" step in the glycolytic pathway • Glucose 6-phosphate and fructose 6-phosphate have other possible fates, but fructose 1,6-bisphosphate is targeted for glycolysis.

• Phosphofructokinase-1 is subject to complex allosteric regulation • The PFK1 has got binding sites for both, ATP-Mg2+ and F6P binding sites • The activity of PFK-1 enzyme and glycolysis is inhibited whenever the cell has high levels of ATP.

In this way, the cell can increase or decrease the rate of glycolysis in response to the cell's energy requirements • In Eukaryotes, fructose 2,6-bisphosphate (not to be confused with the PFK-1 reaction product, fructose 1,6-bisphosphate) is a potent allosteric activator of PFK-1

• Ribulose 5-phosphate, an intermediate in the pentose phosphate pathway also activates phosphofructokinase indirectly.

✓ Define Gluconeogenesis?

The process of synthesis of glucose from non-carbohydrate sources is known as Gluconeogenesis (abbreviated GNG) (“new formation of sugar”),

OR

Gluconeogenesis is the pathway, which converts pyruvate and related noncarbohydrate carbon compounds to glucose.

✓ Ping pong mechanism

Ping-pong mechanism of nucleoside diphosphate kinase:

The enzyme binds its first substrate (ATP in our example), and a phosphoryl group is transferred to the side chain of a His residue. ADP departs, and another nucleoside (or deoxynucleoside) diphosphate replaces it, and this is converted to the corresponding triphosphate by transfer of the phosphoryl group from the phosphohistidine residue.

This reaction is fully reversible, so after the intense demand for ATP ends, the enzyme can recycle AMP by converting it to ADP which can then be phosphorylated to ATP in mitochondria

✓ Write equation of glycolysis? (3)

Glucose + 2NAD+ + 2ADP + 2Pi -------→ 2 pyruvate + 2NADH + 2H+ + 2ATP + 2H2O

✓ What snRNA? (3)

Small nuclear RNA (snRNA)

• Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells. • Most of them are complexed with proteins to form ribonucleoproteins. Nucleotides and Nucleic Acids

• They are distributed in the nucleus, in the cytoplasm or in both.

• They are significantly involved in rRNA and mRNA processing and gene regulation.

✓ What enzymes used in brewer and baker reaction? 3

• Pyruvate decarboxylase is present in brewer's and baker's yeast (**Saccharomyces cerevisiae**) and in all other organisms that ferment glucose to ethanol, including some plants.

✓ Factor affecting melting temperature of DNA.

• **Tm:**

When DNA is heated, the temperature at which one half of the helical structure is lost is defined as the melting temperature.

• A specific sequence of DNA has a characteristic denaturation temperature, or melting point (Tm) • Careful determination of the melting temperature of a DNA specimen, under fixed conditions of pH and ionic strength, can yield an estimate of a DNA base composition.

✓ Describe primary structure of tRNA

• tRNA molecules consist of 74-95 nucleotides in a particular sequence.

• The t RNA molecules contain not only the usual bases like adenine, guanine, cytosine, uracil but also contain unusual bases

• These unusual bases(also called modified bases) include

• Dihydrouracil

• Pseudouridine

• Thymine.

✓ Laws of thermodynamics

• There are two fundamental laws of thermodynamics.

1. First law The first law is the principle of the conservation of energy: it states that "for any physical or chemical change, the total amount of energy in the universe remains constant;

energy may change form or it may be transported from one region to another, but it cannot be created or destroyed."

2. Second law The second law of thermodynamics states that
   "The universe always tends towards increasing disorder and in all natural processes, the entropy of the universe increases… give example of hot cup.. Now the point is that entropy is not a measure of energy. It only describes that how energy is distributed within a system

✓ Bisphosphates

Bisphosphates

Compounds that contain two phosphate or phosphoryl groups attached at different positions in the molecule are named bisphosphates (or bisphospho compounds); for example, fructose 1,6-bisphosphate and 1,3-bisphosphoglycerate.

Diphosphates

Compounds with two phosphates linked together as a pyrophosphoryl group are named diphosphates; for example, adenosine diphosphate (ADP).

Triphosphate

Similar rules apply for the naming of diphosphate (such as inositol 1,4,5-trisphosphate;) and triphosphates (such as adenosine triphosphate, ATP)

✓ Advantages of fermentation?

Fermentation is a metabolic process that converts sugars to acids, gases, or alcohol only in the absence of oxygen. It occurs in yeast and bacteria, and also in oxygen-starved muscle cells, as in the case of lactic acid fermentation.

Fermentation is also used more broadly to refer to the bulk growth of microorganisms on a growth medium, often with the goal of producing a specific chemical product.

Fermentations are carried out by a wide range of organisms, many of which occupy anaerobic niches, and they yield a variety of end products, some of which are used commercially.

✓ Elimination reaction

• The loss of water from an alcohol, resulting in the introduction of a C=C bond b/w 2 adjacent carbon atoms • An elimination reaction with no changes in oxidation state

✓ Glucose

Glucose is a simple sugar with the molecular formula $C_6H_{12}O_6$, which means that it is a molecule that is made of six carbon atoms, twelve hydrogen atoms, and six oxygen atoms. Glucose circulates in the blood of animals as blood sugar.

Long Questions:

(These are Most Repeated Questions)

✓ Importance of glycogenesis

The primary purpose of glycogenesis is **to make sure the body does not run out of glucose**. Glucose is important as it is the body's primary source of energy. Without an adequate supply of glucose in the body, vital organs will eventually shut down.

Gluconeogenesis refers to synthesis of new glucose from noncarbohydrate precursors, **provides glucose when dietary intake is insufficient or absent**. It also is essential in the regulation of acid-base balance, amino acid metabolism, and synthesis of carbohydrate derived structural components.

Glucose is the only energy source used by the brain (with the exception of ketone bodies during times of fasting), testes, erythrocytes, and kidney medulla.

✓ Features of denaturing DNA in laboratory

DNA Denaturation

• Separation of the two strands of the double helix when hydrogen bonds between the paired bases are disrupted.

Disruption can occur in the laboratory if

• The pH or the salt concentration of the DNA solution is altered

• If the solution is heated above 80°C

• Disruption of the hydrogen bonds between paired bases and of base stacking causes unwinding of the double helix to form two single strands complete separation of DNA stands along the entire length.

• No covalent bonds in the DNA are broken

✓ snRNA 5marks

Small nuclear RNA (snRNA)

• Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells. • Most of them are complexed with proteins to form ribonucleoproteins.

• They are distributed in the nucleus, in the cytoplasm or in both. • They are significantly involved in rRNA and mRNA processing and gene regulation.

✓ Types of biochemical reaction in living system

The reactions going on in biological systems can be categorized into 5 types:

1. Reactions that make or break carbon-carbon bonds

2. Internal rearrangements, isomerizations, and eliminations

3. Free-radical reactions

4. Group transfer

5. Oxidation-reduction reactions

✓ Discuss 5 disease which occurs due to ion defects

Protein Energy Malnutrition, Scurvy, Rickets, Beriberi, Hypocalcemia, Osteomalacia, Vitamin K Deficiency, Pellagra, Xerophthalmia, and Iron Deficiency.

✓ Write the steps of synthesis of phosphoenolpyruvate into pyruvate ...

Conversion of Pyruvate to Phosphoenolpyruvate

• The first of the bypass reactions in gluconeogenesis is the conversion of pyruvate to phosphoenolpyruvate (PEP). • This reaction cannot occur by simple reversal of the pyruvate kinase reaction • Instead, the phosphorylation of pyruvate is achieved by a sequence of reactions that in eukaryotes requires 2 enzymes in both the cytosol and mitochondria.

• Pyruvate is first transported from the cytosol into mitochondria or is generated from alanine within mitochondria by transamination, in which the alpha -amino group is transferred from alanine (leaving pyruvate) to an alpha-keto carboxylic acid

• In mitochondria, pyruvate is converted to oxaloacetate in a biotin (coenzyme)-requiring reaction catalyzed by pyruvate carboxylase, a mitochondrial enzyme

Alternative paths from pyruvate to phosphoenolpyruvate

• Oxalo acetic acid (OAA) produced in mitochondria is impermeable through mitochondrial membrane Oxaloacetate + NADH + H+ ←---→ L-malate + NAD+

• Therefore, Oxalo acetic acid is converted into malate, exported from the mitochondrion into the cytoplasm, converted back to oxaloacetate inorder to allow gluconeogenesis

✓ Cytochromes roles in plant growth and mammalian circadian rhythms?? (5)

• Family of flavoproteins,

• Found in both bacteria and eukaryotes

• mediate the effects of blue light on plant development and the effects of light on mammalian circadian rhythms (oscillations in physiology and biochemistry, with a 24-hour period)

✓ NcRNA?

Noncoding Regulatory RNAs

• One of the most exciting discoveries in the last decade of eukaryotic regulatory biology has been the identification and characterization of regulatory nonprotein coding RNAs (ncRNAs).

• NcRNAs exist in two general size classes,

1) small consisting of microRNA (miRNAs) and silencing (siRNAs)
2) Large consisting of long noncoding RNAs (lncRNAs)

• The small ncRNAs termed microRNA (miRNAs) and silencing (siRNAs) typically inhibit gene expression at the level of specific protein production by targeting mRNAs through one of several distinct mechanisms.

Both siRNAs and miRNAs typically hybridize, via the formation of RNA–RNA hybridization to their targeted mRNAs

✓ Define oxidation and reduction reactions?

Oxidation	Reduction
Addition of oxygen	Removal of oxygen
loss of electrons	Gain of electrons
Loss of hydrogen	Gain of hydrogen

✓ Write down the 5 steps of oxidative phase of pentose phosphate pathway?

1. The Oxidative Phase Produces Pentose Phosphates and NADPH

2. The first reaction of the pentose phosphate pathway is the oxidation of glucose 6-phosphate by glucose 6- phosphate dehydrogenase (G6PD) to form 6- phosphoglucono-delta-lactone, an intramolecular ester.

3. NADP+ is the electron acceptor, and overall equilibrium is in the direction of NADPH formation.

4. The lactone is hydrolyzed to the free acid 6- phosphogluconate by a specific enzyme lactonase, then 6-phosphogluconate undergoes oxidation and decarboxylation by 6-phosphogluconate dehydrogenase

• The reaction generates a second molecule of NADPH. Phosphopentose isomerase converts ribulose 5- phosphate to its aldose isomer, ribose 5-phosphate.

In some tissues, the pentose phosphate pathway ends at this point, and its overall equation for reactions is

G6-P + 2NADP+ + H2O -------→ ribulose 5-phosphate + CO2 + 2NADPH + 2H +

• The net result is the production of NADPH, a reducing agent for biosynthetic reactions, and ribose 5- phosphate, a precursor for nucleotide synthesis

✓ Define medium? Write their Characteristics?

“A medium which is used for a large scale fermentation, in order to ensure the sustainability of the operation”

1. It should be cheap and easily available

2. It should maximize the growth of the microorganism, productivity and the rate of formation of the desired product

3. It should minimize the formation of undesired products

✓ Write down types of RNA? And explain snRNA? (5)

(i)Messenger RNAs (mRNAs).

(ii) Ribosomal RNAs (rRNAs)

(iii) Transfer RNAs (tRNAs)

(iv) Small nuclear RNA (snRNA

Small nuclear RNA (snRNA)

Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells. Most of them are complexed with proteins to form ribonucleoproteins. Nucleotides and Nucleic Acids

They are distributed in the nucleus, in the cytoplasm or in both. They are significantly involved in rRNA and mRNA processing and gene regulation.

✓ Beaker yeast

• Pyruvate decarboxylase is present in brewer's and baker's yeast (Saccharomyces cerevisiae) and in all other organisms that ferment glucose to ethanol, including some plants.

• The CO2 produced by pyruvate decarboxylation in brewer's yeast is responsible for the characteristic carbonation of champagne (ethanol fermentation). • In baking, CO2 released by pyruvate decarboxylase when yeast is mixed with a fermentable sugar causes dough to rise. • The enzyme is absent in organisms that carry out lactic acid fermentation.

✓ Explain fermentation and it's types along with intermediate products and enzymes...

Fermentation actually comes from Latin word "fervere" which means "to boil"

• So the word "Fermentation" is referred to physical state of boiling/bubbling • The bubbling appearance in fermentation is due to the production of CO2 bubbles caused by anaerobic catabolism of sugars • Fermentation is a metabolic process that converts sugars to acids, gases, or alcohol

• The science of fermentation is known as ZYMOLOGY

Fermentation types:

lactic acid fermentation, alcohol fermentation, acetic acid fermentation, and butyric acid fermentation.

Products:

Common fermented foods include kimchi, sauerkraut, kefir, tempeh, kombucha, and yogurt. These foods may reduce heart disease risk and aid digestion, immunity, and weight loss.

✓ Types of chemical transformation during glycolysis

In the sequential reactions of glycolysis, three types of chemical transformations are noteworthy:

(1) Degradation of the carbon skeleton of glucose to yield pyruvate

(2) Phosphorylation of ADP to ATP by compounds with high phosphoryl group transfer potential, formed during glycolysis

(3) Transfer of a hydride ion to NAD+ forming NADH.

✓ Name six bioreactors/ Fermentor

Types of Bioreactors

• Simple fermenters (batch and continuous)

• Fed batch fermenter

• Air-lift or bubble fermenter

• Cyclone column fermenter

• Tower fermenter

• Packed bed bioreactor

• photobioreacto

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